

Nộp bài tập homework ss08

Tên	Hoàng Duy Hương
Lớp	PD16

- 2 Path timing đề bài yêu cầu report

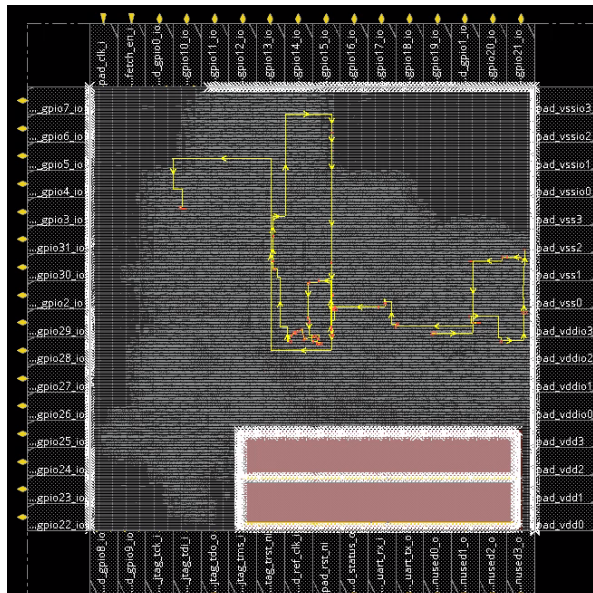
```
2. report timing for these 2 timing paths:  
2.1 report_timing -from i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_i_controller_i_instr_i_21__reg -to i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_iFIFOMem_326__reg  
2.2 report_timing -from i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_i_controller_i_instr_i_15__reg -to i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_iFIFOMem_366__reg
```

- Report timing path 2.1 và kết quả là hình dưới

```
Path 1: VIOLATED Setup Check with Pin i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_iFIFOMem_326__reg/CLK  
Endpoint: i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_iFIFOMem_326__reg/D (*) checked with leading edge of 'clk_sys'  
Beginpoint: i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_i_controller_i_instr_i_21__reg/Q (*) triggered by leading edge of 'clk_sys'  
Path Groups: {reg2reg}  
Analysis View: func_view_wc  
Other End Arrival Time 0.000  
- Setup 0.164  
+ Phase Shift 10.000  
+ CPBR Adjustment 0.000  
- Uncertainty 0.100  
= Required Time 9.736  
- Arrival Time 10.095  
= Slack Time -0.359  
Clock Rise Edge 0.000  
+ Clock Network Latency (Ideal) -0.300  
= Beginpoint Arrival Time -0.300
```

Pin	Cell	Fanout	Load	Slew	Delay	Arrival Time	Pin Location
i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_i_controller_i_instr_i_21__reg/CLK ->	-	4983	0.000	0.200	-	-0.300	(1172.16, 749.28)
i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_i_controller_i_instr_i_21__reg/Q	sg13g2_dfrbp_1	1	0.009	0.074	0.424	0.124	(1162.56, 750.12)
i_croc_soc/i_croc/i_core_wrap/FE_0FC3423_i_ibex_id_stage_i_controller_i_instr_i_21/X	sg13g2_buf_8	6	0.074	0.079	0.158	0.282	(1164.00, 744.24)
i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_i_4984/Y	sg13g2_nor2b_2	1	0.010	0.095	0.203	0.486	(1283.04, 780.36)
i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_i/FE_OCPC5730_1910/X	sg13g2_buf_8	2	0.054	0.063	0.160	0.645	(1285.44, 782.04)
i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_i/FE_OCPC4261_FE_OFN1016_1910/Y	sg13g2_inv_16	6	0.078	0.043	0.060	0.706	(1295.04, 779.10)
i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_i/FE_OCPC4180_FE_OFN1016_1910/Y	sg13g2_inv_8	8	0.069	0.069	0.072	0.777	(1269.60, 797.16)
i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_i/FE_OCPC4186_FE_OFN1016_1910/X	sg13g2_buf_8	7	0.070	0.073	0.161	0.938	(1363.20, 727.44)
i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_i/FE_OCPC4188_FE_OFN1016_1910/X	sg13g2_buf_16	8	0.064	0.049	0.141	1.079	(1425.12, 808.92)
i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_i/FE_RC_402_0/Y	sg13g2_a22oi_1	1	0.011	0.160	0.153	1.232	(1430.40, 990.36)
i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_i/FE_RC_287_1/Y	sg13g2_a21oi_2	1	0.010	0.124	0.158	1.390	(1419.84, 963.90)
i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_i/FE_RC_6232_0/Y	sg13g2_inv_1	1	0.005	0.055	0.086	1.476	(1380.76, 968.32)
i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_i/FE_RC_6231_0/X	sg13g2_nand2_2	1	0.021	0.078	0.170	1.655	(1366.56, 969.36)
i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_i/FE_RC_283_0/Y	sg13g2_nand2_2	1	0.009	0.052	0.073	1.728	(1272.48, 957.60)
i_croc_soc/i_croc/i_core_wrap/FE_OCPC4278_i_ibex_ex_block_i_multdiv_operand_b_i_17/X	sg13g2_buf_8	2	0.076	0.082	0.145	1.873	(1276.80, 955.92)
i_croc_soc/i_croc/i_core_wrap/FE_RC_4814_0/Y	sg13g2_o21ai_1	1	0.008	0.120	0.152	2.025	(1059.36, 766.92)
i_croc_soc/i_croc/i_core_wrap/FE_RC_4812_0/Y	sg13g2_nor2b_2	1	0.022	0.172	0.184	2.209	(1062.72, 774.48)
i_croc_soc/i_croc/i_core_wrap/FE_OCPC6426_FE_RN_2722_0/Y	sg13g2_inv_8	6	0.061	0.079	0.108	2.317	(1065.12, 774.48)
i_croc_soc/i_croc/i_core_wrap/_5466_dup/Y	sg13g2_nand2b_1	2	0.010	0.118	0.197	2.514	(1029.12, 846.30)
i_croc_soc/i_croc/i_core_wrap/_5469_Y	sg13g2_nor2_2	1	0.007	0.085	0.111	2.625	(1028.64, 834.96)
i_croc_soc/i_croc/i_core_wrap/FE_RC_8227_0/Y	sg13g2_nor2_2	1	0.008	0.049	0.081	2.706	(1030.56, 834.96)

- Highlight path trên GUI



- Path này bị violate setup timing
- Có slack violate là -0.359
- Em sẽ tiến hành tìm kiếm các cell nào có delay lớn và thử upsize cell lên.
- Tiến hành bật batchmode để upsize cell theo như hướng dẫn ss07

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innovus (batch mode) 60> history
41 ecoChangeCell -inst i_croc_soc/i_croc/i_uart/FE_ECOC8549_FE_OFN5467_FE_OFN2310_all_periph_obi_req_233 -cell sg13g2_buf_4
42 report_timing -from i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_i_instr_i_21_reg -to i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_iFIFOMem_326__reg > report_timing_1.rpt
43 ecoChangeCell -inst i_croc_soc/i_croc/i_uart/FE_ECOC8549_FE_OFN5467_FE_OFN2310_all_periph_obi_req_233 -cell sg13g2_buf_8
44 report_timing -from i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_i_instr_i_21_reg -to i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_iFIFOMem_326__reg > report_timing_1.rpt
45 ecoChangeCell -inst i_croc_soc/i_croc/i_uart/FE_ECOC8549_FE_OFN5467_FE_OFN2310_all_periph_obi_req_233 -cell sg13g2_buf_16
46 report_timing -from i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_i_instr_i_21_reg -to i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_iFIFOMem_326__reg > report_timing_1.rpt
47 ecoChangeCell -inst i_croc_soc/i_croc/i_uart/FE_ECOC8549_FE_OFN5467_FE_OFN2310_all_periph_obi_req_233 -cell sg13g2_buf_4
48 report_timing -from i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_i_instr_i_21_reg -to i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_iFIFOMem_326__reg > report_timing_1.rpt
49 ecoChangeCell -inst i_croc_soc/i_croc/i_uart/FE_ECOC8549_FE_OFN5467_FE_OFN2310_all_periph_obi_req_233 -cell sg13g2_buf_8
50 report_timing -from i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_i_instr_i_21_reg -to i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_iFIFOMem_326__reg > report_timing_1.rpt
51 ecoChangeCell -inst i_croc_soc/i_croc/i_uart/FE_ECOC8552_FE_OFN5467_FE_OFN2310_all_periph_obi_req_233 -cell sg13g2_buf_4
52 report_timing -from i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_i_instr_i_21_reg -to i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_iFIFOMem_326__reg > report_timing_1.rpt
53 ecoChangeCell -inst i_croc_soc/i_croc/i_uart/FE_ECOC8552_FE_OFN5467_FE_OFN2310_all_periph_obi_req_233 -cell sg13g2_buf_8
54 report_timing -from i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_i_instr_i_21_reg -to i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_iFIFOMem_326__reg > report_timing_1.rpt
55 ecoChangeCell -inst i_croc_soc/i_croc/i_uart/FE_ECOC1_FE_OFN5467_FE_OFN2310_all_periph_obi_req_233 -cell sg13g2_buf_8
56 report_timing -from i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_i_instr_i_21_reg -to i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_iFIFOMem_326__reg > report_timing_1.rpt
57 ecoChangeCell -inst i_croc_soc/i_croc/i_uart/FE_ECOC0_FE_OFN5467_FE_OFN2310_all_periph_obi_req_233 -cell sg13g2_buf_8
58 report_timing -from i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_i_instr_i_21_reg -to i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_iFIFOMem_326__reg > report_timing_1.rpt
59 clear
60 history
innovus (batch mode) 61>

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- Do đoạn code trước dòng 41 em clean mà quên cap lại hisory nên chỉ còn đoạn sau á!
- Sau khi đã upsize các cell inv/buf có delay lớn thì path timing đã MET setup timing

```

1 #####
2 # Generated by: Cadence Innovus 20.10-p0804.1
3 # OS: Linux x86_64 (Host ID: ietc-eda-be-9-ldap-1)
4 # Generated on: Tue Aug 4 20:50:22 2020
5 # Design: croc_chip
6 # Command: report_timing -from i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_i_instr_i_21_reg -to i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_iFIFOMem_326__reg > report_timing_1.rpt
7 #####
8 Path 1: MET Setup Check with Pin i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_iFIFOMem_326__reg/CLK
9 Endpoint: i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_iFIFOMem_326__reg/D (*) checked with leading edge of 'clk_sys'
10 Beginpoint: i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_i_instr_i_21_reg/Q (*) triggered by leading edge of 'clk_sys'
11 Path Groups: (reg2reg)
12 Analysis View: func_view_wc
13 Other End Arrival Time 0.000
14 - Setup 0.164
15 + Phase Shift 10.000
16 + CPRP Adjustment 0.000
17 - Uncertainty 0.100
18 = Required Time 9.726
19 - Arrival Time 9.632
20 = Slack Time 0.104
21 Clock Rise Edge 0.000
22 + Clock Network Latency (Ideal) -0.300
23 = Beginpoint Arrival Time -0.300
24 -----
25 Pin Cell Fanout Load Slew Delay Arrival Time Pin Location
26 -----
27 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_i_instr_i_21_reg/CLK - 4983 0.000 0.200 - -0.300 (1172.16, 740.28)
28 sg13g2_dfrbp_1 1 0.009 0.074 0.424 0.124 (1162.56, 750.12)
29 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_i_instr_i_21/X sg13g2_buf_8 6 0.074 0.079 0.158 0.282 (1164.00, 744.24)
30 sg13g2_nor2b_2 1 0.010 0.095 0.203 0.486 (1283.04, 780.36)
31 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_OCPC57380_1910/X sg13g2_buf_8 2 0.054 0.063 0.100 0.645 (1285.44, 782.04)
32 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_OCPC4261_FE_OFN1016_1910/Y sg13g2_inv_16 6 0.078 0.043 0.060 0.706 (1295.04, 779.10)
33 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_OCPC4180_FE_OFN1016_1910/Y sg13g2_inv_8 8 0.069 0.069 0.072 0.777 (1269.60, 797.10)
34 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_OCPC4186_FE_OFN1016_1910/X sg13g2_buf_8 7 0.070 0.073 0.161 0.938 (1363.20, 727.44)
35 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_OCPC4188_FE_OFN1016_1910/X sg13g2_buf_16 8 0.064 0.049 0.141 1.079 (1425.12, 808.92)
36 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_402_0/Y sg13g2_a2201_1 1 0.011 0.160 0.153 1.232 (1430.40, 990.36)
37 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_287_1/Y sg13g2_a2101_2 1 0.010 0.124 0.158 1.390 (1419.04, 963.00)
38 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_6232_0/Y sg13g2_inv_1 1 0.005 0.055 0.086 1.476 (1380.76, 968.32)
39 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_6231_0/X sg13g2_and2_2 1 0.021 0.078 0.179 1.655 (1366.56, 969.36)
40 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_283_0/Y sg13g2_nand2_2 1 0.009 0.052 0.073 1.728 (1272.48, 957.60)
41 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_402_0/Y sg13g2_buf_8 2 0.076 0.082 0.145 1.873 (1276.80, 955.92)
42 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_287_1/Y sg13g2_a2101_1 1 0.008 0.120 0.152 2.025 (1059.36, 766.92)
43 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_4812_0/Y sg13g2_nor2b_2 1 0.022 0.172 0.184 2.209 (1062.72, 774.48)
44 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_6231_0/X sg13g2_inv_8 6 0.061 0.079 0.108 2.317 (1065.12, 774.48)
45 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_5466_dup/Y sg13g2_nand2b_2 2 0.010 0.077 0.206 2.523 (1029.60, 846.72)
46 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_5469/Y sg13g2_nor2_2 1 0.007 0.078 0.097 2.620 (1028.64, 834.96)
47 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_8227_0/Y sg13g2_nor2_2 1 0.008 0.047 0.078 2.690 (1030.56, 834.96)
48

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- Report timing path 2.2 và kết quả là hình dưới

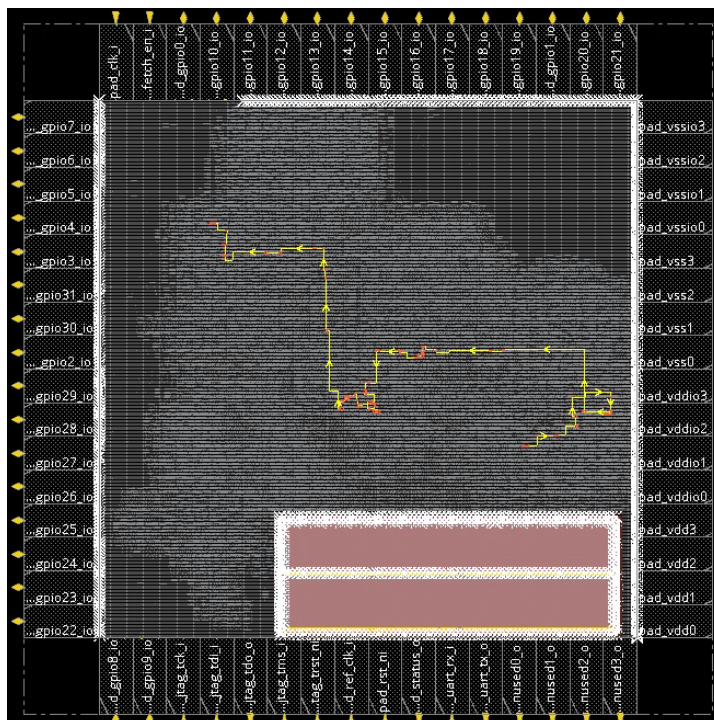
```

Path 1: VIOLATED Setup Check with Pin i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_1IFIFOMem_366_reg/CLK
Endpoint: i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_1IFIFOMem_366_reg/D (*) checked with leading edge of 'clk_sys'
Beginpoint: i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15_reg/Q (*) triggered by leading edge of 'clk_sys'
Path Groups: {reg2reg}
Analysis View: func_view_wc
Other End Arrival Time 0.000
+ Setup 0.164
+ Phase Shift 10.000
+ CPBR Adjustment 0.000
+ Uncertainty 0.100
+ Required Time 9.736
+ Arrival Time 9.924
+ Slack Time -0.187
+ Clock Rise Edge 0.000
+ Clock Network Latency (Ideal) -0.300
= Beginpoint Arrival Time -0.300

```

Pin	Cell	Fanout	Load	Slew	Delay	Arrival Time	Pin Location
i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15_reg/CLK ->	-	4983	0.000	0.200	-	-0.300	(1188.00, 635.88)
i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15_reg/Q	sg13g2_dfrbp_1	1	0.011	0.009	0.435	0.135	(1197.60, 636.72)
i_croc_soc/i_croc/i_core_wrap/FE_OCPC3767_1_ibex_id_stage_1_controller_1_instr_1_15/Y	sg13g2_inv_4	4	0.044	0.075	0.099	0.234	(1199.04, 636.72)
i_croc_soc/i_croc/i_core_wrap/FE_OCPC3771_1_ibex_id_stage_1_controller_1_instr_1_15/Y	sg13g2_inv_2	2	0.062	0.067	0.080	0.315	(1220.16, 661.08)
i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_OCPC3842_FE_OFN572_1_ibex_id_stage_1_controller_1_instr_1_15/Y	sg13g2_inv_16	4	0.088	0.047	0.067	0.381	(1263.84, 662.34)
i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_OCPC3845_FE_OFN572_1_ibex_id_stage_1_controller_1_instr_1_15/Y	sg13g2_inv_8	2	0.032	0.040	0.057	0.438	(1316.64, 683.76)
i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_OCPC3780_1_ibex_id_stage_1_controller_1_instr_1_15/Y	sg13g2_inv_8	2	0.019	0.025	0.041	0.479	(1312.32, 706.44)
i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_OCPC3786_1_ibex_id_stage_1_controller_1_instr_1_15/Y	sg13g2_inv_2	8	0.104	0.325	0.240	0.718	(1305.12, 757.68)
i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_OCPC3789_1_ibex_id_stage_1_controller_1_instr_1_15/Y	sg13g2_inv_2	1	0.018	0.111	0.172	0.890	(1398.24, 751.80)
i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_OCPC306_1_ibex_id_stage_1_controller_1_instr_1_15/Y	sg13g2_inv_4	3	0.035	0.072	0.096	0.956	(1389.60, 714.00)
i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/4608_JX	sg13g2_mux4_1	1	0.004	0.068	0.361	1.347	(1392.48, 712.32)
i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_491_0/Y	sg13g2_nand2_1	1	0.007	0.069	0.081	1.429	(1393.92, 711.90)
i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_490_0/Y	sg13g2_nand2_2	1	0.007	0.057	0.092	1.521	(1398.24, 718.20)
i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_489_0/Y	sg13g2_nor2_2	1	0.013	0.117	0.133	1.654	(1396.32, 721.56)
i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_9237_0/Y	sg13g2_nand3_1	1	0.008	0.164	0.188	1.842	(1331.52, 718.20)
i_croc_soc/i_croc/i_core_wrap/FE_OCPC6571_1_ibex_ex_block_1_multdiv_operand_a_1_6/X	sg13g2_buf_8	1	0.060	0.066	0.197	2.040	(1334.88, 719.88)
i_croc_soc/i_croc/i_core_wrap/FE_RC_222_0/Y	sg13g2_nand2_2	1	0.006	0.047	0.067	2.107	(1141.92, 866.88)
i_croc_soc/i_croc/i_core_wrap/FE_RC_221_1/Y	sg13g2_nand2_2	1	0.011	0.081	0.091	2.198	(1139.04, 866.88)
i_croc_soc/i_croc/i_core_wrap/FE_RC_219_0/Y	sg13g2_nand2_2	1	0.011	0.112	0.136	2.336	(1111.20, 863.10)
i_croc_soc/i_croc/i_core_wrap/FE_RC_220_1/Y	sg13g2_inv_4	6	0.057	0.096	0.122	2.459	(1107.84, 863.52)
i_croc_soc/i_croc/i_core_wrap/_5627_Y	sg13g2_nor2_2	2	0.007	0.079	0.108	2.567	(981.60, 863.52)
i_croc_soc/i_croc/i_core_wrap/_6194_Y	sg13g2_nand2b_1	4	0.023	0.166	0.209	2.776	(974.88, 867.30)
i_croc_soc/i_croc/i_core_wrap/_6195_Y	sg13g2_inv_1	2	0.011	0.092	0.136	2.913	(952.60, 873.81)
i_croc_soc/i_croc/i_core_wrap/FE_RC_212_0/Y	sg13g2_nor2_1	1	0.005	0.106	0.123	3.036	(949.92, 871.08)
i_croc_soc/i_croc/i_core_wrap/FE_RC_210_0/Y	sg13g2_nand2_1	1	0.003	0.068	0.097	3.133	(948.48, 858.06)
i_croc_soc/i_croc/i_core_wrap/FE_RC_211_0/Y	sg13g2_inv_1	3	0.015	0.109	0.118	3.251	(952.04, 858.06)
i_croc_soc/i_croc/i_core_wrap/FE_RC_8168_0/Y	sg13g2_nand3_1	2	0.010	0.182	0.215	3.465	(952.32, 854.28)

- Path này cũng violate setup timing và có slack là -0.187
- Yêu cầu đề bài fix path này bằng 2 cách
 - Up/down size cell
 - Check net detour hoặc loại bỏ các buffer dư thừa
- Đầu tiên cách 1 em vẫn bật batchmode lên và upsize các cell có delay lớn
- Highlight path này trên GUI



- Sau khi fix thì path đã MET setup timing với slack dương như hình bên dưới

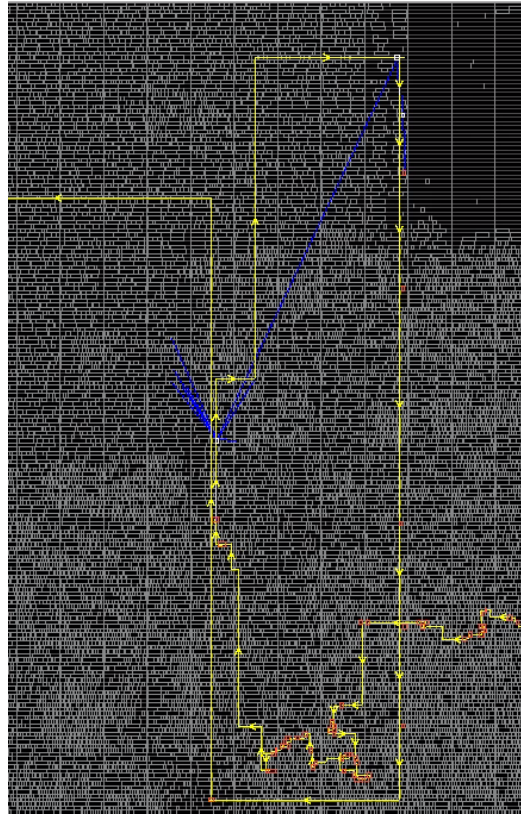
```
innovus (batch mode) 115> history
96 ecochangeCell -inst i_croc_soc/i_croc/FE_OPC6136_1171 -cell sg13g2_buf_8
97 report_timing -from i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15_reg -to i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_iFIFOMem_366_reg > report_timing_2.rpt
98 dget head.1bCells.name sg13g2_mux2_4
99 ecochangeCell -inst i_croc_soc/i_croc/i_uart/FE_OFC3179_02995 -cell sg13g2_buf_16
100 report_timing -from i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15_reg -to i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_iFIFOMem_366_reg > report_timing_2.rpt
101 ecochangeCell -inst i_croc_soc/i_croc/i_uart/FE_OFC3179_02995 -cell sg13g2_buf_8
102 report_timing -from i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15_reg -to i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_iFIFOMem_366_reg > report_timing_2.rpt
103 ecochangeCell -inst i_croc_soc/i_croc/i_uart/FE_OFC2532_02999 -cell sg13g2_buf_8
104 report_timing -from i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15_reg -to i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_iFIFOMem_366_reg > report_timing_2.rpt
105 ecochangeCell -inst i_croc_soc/i_croc/i_uart/FE_OFC2532_02999 -cell sg13g2_buf_4
106 report_timing -from i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15_reg -to i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_iFIFOMem_366_reg > report_timing_2.rpt
107 dget head.1bCells.name sg13g2_mux2_4
108 ecochangeCell -inst i_croc_soc/i_croc/i_uart/_06423_ -cell sg13g2_mux2_2
109 report_timing -from i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15_reg -to i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_iFIFOMem_366_reg > report_timing_2.rpt
110 ecochangeCell -inst i_croc_soc/i_croc/i_uart/_06423_ -cell sg13g2_mux2_1
111 report_timing -from i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15_reg -to i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_iFIFOMem_366_reg > report_timing_2.rpt
112 dget head.1bCells.name sg13g2_mux4_4
113 ecochangeCell -inst i_croc_soc/i_croc/i_core_wrap/FE_OPC6571_1_ibex_ex_block_1_multdiv_operand_a_1_6 -cell sg13g2_buf_4
114 report_timing -from i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15_reg -to i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_iFIFOMem_366_reg > report_timing_2.rpt
115 history
innovus (batch mode) 116>
```

```
1 #####
2 # Generated by: Cadence Innovus 20.10-p084.1
3 # OS: Linux x86_64(Host ID tctc-eda-be-9-ldap-1)
4 # Generated on: Tue Aug 4 21:37:53 2026
5 # Design: croc_chip
6 # Command: report_timing -from i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15_reg -to i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_iFIFOMem_366_reg > report_timing_2.rpt
7 #####
8 Path 1: MET Setup Check with Pin i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_iFIFOMem_366_reg/CLK
9 Endpoints: i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_iFIFOMem_366_reg/0 (*) checked with leading edge of 'clk_sys'
10 Beginpoint: i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15_reg/Q (*) triggered by leading edge of 'clk_sys'
11 Path Groups: {reg2reg}
12 Analysis View: func_view_wc
13 Other End Arrival Time
14 - Setup 0.164
15 + Phase Shift 10.000
16 + CPRP Adjustment 0.000
17 - Uncertainty 0.100
18 + Required Time 9.736
19 - Arrival Time 9.822
20 - Slack Time 0.005
21 Clock Rise Edge 0.000
22 + Clock Network Latency (Ideal) -0.300
23 = Beginpoint Arrival Time -0.300
24
25 Pin Cell Fanout Load Slew Delay Arrival Time Pin Location
26
27
28 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15_reg/CLK -> - 4983 0.000 0.200 - -0.300 (1188.06, 635.88)
29 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15_reg/Q sg13g2_dfrbp_1 1 0.011 0.089 0.435 0.135 (1197.60, 636.72)
30 i_croc_soc/i_croc/i_core_wrap/FE_OPC3767_1_ibex_id_stage_1_controller_1_instr_1_15/Y sg13g2_inv_4 4 0.044 0.075 0.099 0.234 (1199.04, 636.72)
31 i_croc_soc/i_croc/i_core_wrap/FE_OPC3771_1_ibex_id_stage_1_controller_1_instr_1_15/Y sg13g2_inv_8 2 0.062 0.067 0.080 0.315 (1220.16, 661.08)
32 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_OPC3842_FE_OFN572_1_ibex_id_stage_1_controller_1_instr_1_15/Y sg13g2_inv_16 4 0.088 0.047 0.067 0.381 (1263.84, 662.34)
33 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_OPC3845_FE_OFN572_1_ibex_id_stage_1_controller_1_instr_1_15/Y sg13g2_inv_2 2 0.032 0.040 0.057 0.438 (1316.64, 683.76)
34 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_OPC3780_1_ibex_id_stage_1_controller_1_instr_1_15/Y sg13g2_inv_8 2 0.024 0.028 0.044 0.482 (1312.32, 706.44)
35 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_OPC3786_1_ibex_id_stage_1_controller_1_instr_1_15/Y sg13g2_inv_4 8 0.104 0.169 0.134 0.616 (1305.12, 757.68)
36 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_OPC3788_1_ibex_id_stage_1_controller_1_instr_1_15/Y sg13g2_inv_2 1 0.016 0.075 0.118 0.734 (1316.36, 765.24)
37 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_OPC3791_1_ibex_id_stage_1_controller_1_instr_1_15/Y sg13g2_inv_4 3 0.022 0.053 0.070 0.804 (1336.32, 795.48)
38 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_OPC4611_1_ibex_id_stage_1_controller_1_instr_1_15/X sg13g2_buf_1 6 0.024 0.161 0.199 1.003 (1338.44, 797.24)
39 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_5089_0/Y sg13g2_nand2_1 1 0.004 0.085 0.122 1.125 (1322.40, 795.00)
40 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_5087_0/Y sg13g2_o2lat_1 1 0.005 0.138 0.088 1.213 (1329.12, 795.48)
41 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_5402_0/Y sg13g2_a22o1_1 1 0.008 0.131 0.181 1.394 (1327.68, 801.36)
42 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_4357_0/Y sg13g2_nand2_1 1 0.005 0.068 0.103 1.407 (1325.76, 820.26)
43 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_3702_0/Y sg13g2_nand2b_1 1 0.004 0.060 0.102 1.599 (1325.28, 829.56)
44 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_44_0/Y sg13g2_nand3_1 1 0.004 0.068 0.075 1.674 (1322.40, 836.64)
45 i_croc_soc/i_croc/i_core_wrap/FE_OPC4385_1_ibex_ex_block_1_multdiv_operand_a_1_16/X sg13g2_buf_4 1 0.064 0.122 0.212 1.886 (1324.32, 834.96)
46 i_croc_soc/i_croc/i_core_wrap/FE_RC_4834_0/Y sg13g2_nand2b_2 1 0.008 0.075 0.115 2.061 (1062.24, 922.32)
47 i_croc_soc/i_croc/i_core_wrap/FE_RC_798_0/Y sg13g2_nand2_2 1 0.008 0.062 0.069 2.070 (1052.16, 927.36)
48 i_croc_soc/i_croc/i_core_wrap/_5429_/X sg13g2_nand2_1 1 0.007 0.087 0.106 2.176 (1039.20, 916.02)
```

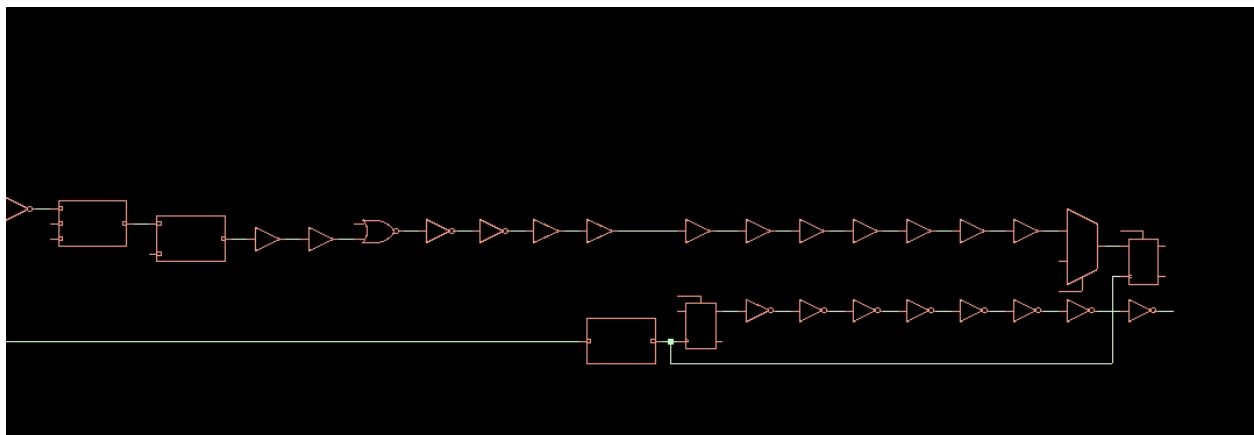
- Em muốn làm thử cách thứ 2 fix timing như đề yêu cầu thì em source lại design như ban đầu thì thấy Slack lại khác với ban đầu, bây giờ violate setup timing slack lại âm đến -0.616 Ps.

```
1 #####
2 # Generated by: Cadence Innovus 20.10-p084.1
3 # OS: Linux x86_64(Host ID tctc-eda-be-9-ldap-1)
4 # Generated on: Tue Aug 4 21:48:18 2026
5 # Design: croc_chip
6 # Command: report_timing -from i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15_reg -to i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_iFIFOMem_366_reg > report_timing_2.rpt
7 #####
8 Path 1: VIOLATED Setup Check with Pin i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_iFIFOMem_366_reg/CLK
9 Endpoints: i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_iFIFOMem_366_reg/0 (*) checked with leading edge of 'clk_sys'
10 Beginpoint: i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15_reg/Q (*) triggered by leading edge of 'clk_sys'
11 Path Groups: {reg2reg}
12 Analysis View: func_view_wc
13 Other End Arrival Time
14 - Setup 0.164
15 + Phase Shift 10.000
16 + CPRP Adjustment 0.000
17 - Uncertainty 0.100
18 + Required Time 9.736
19 - Arrival Time 10.353
20 - Slack Time -0.616
21 Clock Rise Edge 0.000
22 + Clock Network Latency (Ideal) -0.300
23 = Beginpoint Arrival Time -0.300
24
25 Pin Cell Fanout Load Slew Delay Arrival Time Pin Location
26
27
28 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15_reg/CLK -> - 4983 0.000 0.200 - -0.300 (1188.06, 635.88)
29 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15_reg/Q sg13g2_dfrbp_1 1 0.011 0.089 0.435 0.135 (1197.60, 636.72)
30 i_croc_soc/i_croc/i_core_wrap/FE_OPC3767_1_ibex_id_stage_1_controller_1_instr_1_15/Y sg13g2_inv_4 4 0.044 0.075 0.099 0.234 (1199.04, 636.72)
31 i_croc_soc/i_croc/i_core_wrap/FE_OPC3771_1_ibex_id_stage_1_controller_1_instr_1_15/Y sg13g2_inv_8 2 0.062 0.067 0.080 0.315 (1220.16, 661.08)
32 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_OPC3842_FE_OFN572_1_ibex_id_stage_1_controller_1_instr_1_15/Y sg13g2_inv_16 4 0.088 0.047 0.067 0.381 (1263.84, 662.34)
33 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_OPC3845_FE_OFN572_1_ibex_id_stage_1_controller_1_instr_1_15/Y sg13g2_inv_2 2 0.032 0.040 0.057 0.438 (1316.64, 683.76)
34 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_OPC3780_1_ibex_id_stage_1_controller_1_instr_1_15/Y sg13g2_inv_8 2 0.019 0.025 0.041 0.479 (1312.32, 706.44)
35 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_OPC3786_1_ibex_id_stage_1_controller_1_instr_1_15/Y sg13g2_inv_2 8 0.104 0.169 0.134 0.616 (1305.12, 757.68)
36 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_OPC3788_1_ibex_id_stage_1_controller_1_instr_1_15/Y sg13g2_inv_4 3 0.022 0.053 0.070 0.804 (1336.32, 795.48)
37 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_OPC4611_1_ibex_id_stage_1_controller_1_instr_1_15/X sg13g2_buf_1 6 0.024 0.161 0.199 1.003 (1338.44, 797.24)
38 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_5089_0/Y sg13g2_nand2_1 1 0.004 0.085 0.122 1.125 (1322.40, 795.00)
39 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_5087_0/Y sg13g2_o2lat_1 1 0.005 0.138 0.088 1.213 (1329.12, 795.48)
40 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_5402_0/Y sg13g2_a22o1_1 1 0.008 0.131 0.181 1.394 (1327.68, 801.36)
41 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_4357_0/Y sg13g2_nand2_1 1 0.005 0.068 0.103 1.407 (1325.76, 820.26)
42 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_3702_0/Y sg13g2_nand2b_1 1 0.004 0.060 0.102 1.599 (1325.28, 829.56)
43 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_44_0/Y sg13g2_nand3_1 1 0.004 0.068 0.075 1.674 (1322.40, 836.64)
44 i_croc_soc/i_croc/i_core_wrap/FE_OPC4385_1_ibex_ex_block_1_multdiv_operand_a_1_16/X sg13g2_buf_4 1 0.064 0.122 0.212 1.886 (1324.32, 834.96)
45 i_croc_soc/i_croc/i_core_wrap/FE_RC_4834_0/Y sg13g2_nand2b_2 1 0.008 0.075 0.115 2.061 (1062.24, 922.32)
46 i_croc_soc/i_croc/i_core_wrap/FE_RC_798_0/Y sg13g2_nand2_2 1 0.008 0.062 0.069 2.070 (1052.16, 927.36)
47 i_croc_soc/i_croc/i_core_wrap/_5627_/Y sg13g2_nand2_1 2 0.007 0.098 0.122 2.459 (1107.84, 863.52)
48 i_croc_soc/i_croc/i_core_wrap/_5627_/Y sg13g2_nand2_2 2 0.007 0.079 0.108 2.567 (081.06, 863.52)
```

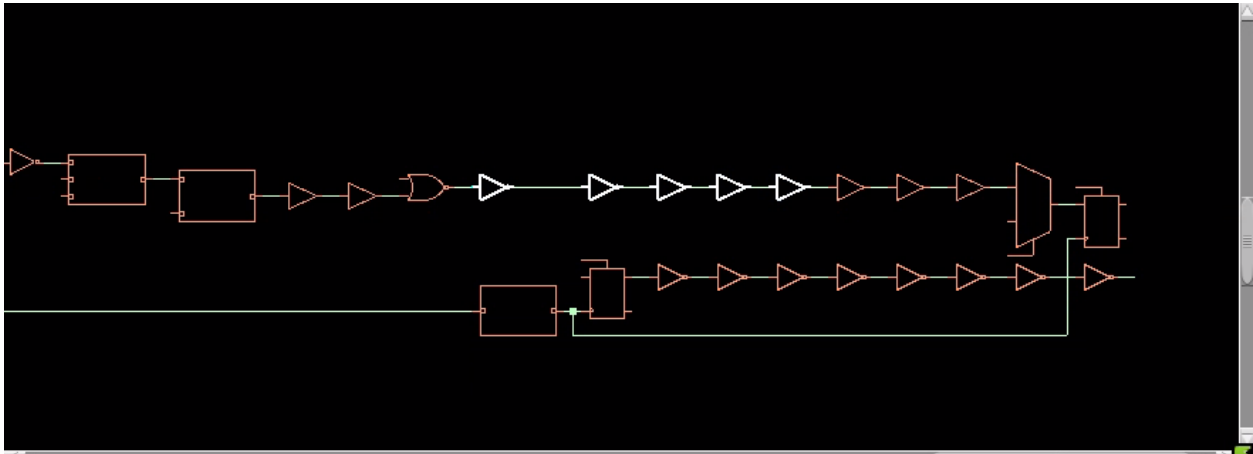
- Theo trên GUI có thể nhận thấy được một cell inv kết nối với một cell buf có kết nối khá xa, dẫn đến delay trên net này khá lớn và có vài buf dư thừa trên đoạn này!



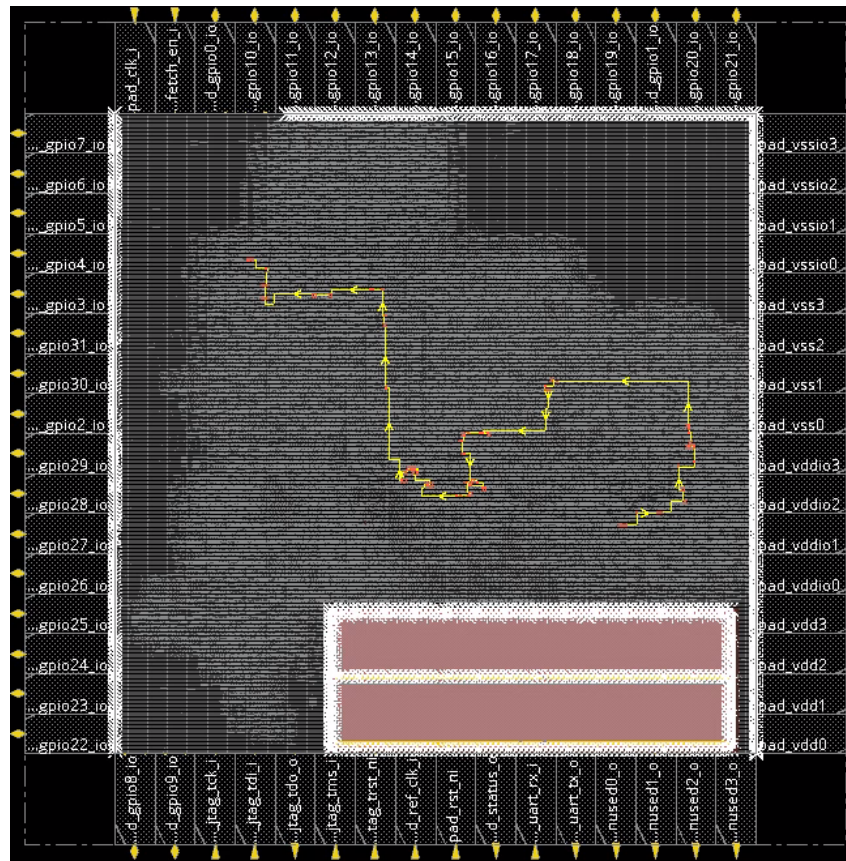
- khi mở timing debug lên thì có thể thấy được đoạn này có khá nhiều buffer thừa, cần select các buffer nào dư thừa và remove các buff thừa đi, sẵn đi lại net bị detour luôn.



- Sau khi remove buffer (bật batchmode và remove theo hướng dẫn ss07)



- Sau khi update timing và đi lại highlight thì sẽ có kết quả như hình dưới đây



- Sau khi tối ưu bằng cách đi lại net và remove buf dư thừa đi thì đã tối ưu khoảng 400 Ps

```
1 #####
2 # Generated by: Cadence Innovus 20.10-p004.1
3 # OS: Linux x86_64 (Host ID: lctc-eda-be-9-ldag-1)
4 # Generated on: Tue Aug 4 22:16:55 2026
5 # Design: croc_chip
6 # Command: report_timing -from i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15_reg/Q -to i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_IFIFOWen_366_reg -report_timing_1.rpt
7 #####
8 Path 1: VIOLATED Setup Check with Pin i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_IFIFOWen_366_reg/CLK
9 Endpoint: i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_IFIFOWen_366_reg/D (*) checked with leading edge of 'clk_sys'
10 Beginpoint: i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15_reg/Q (*) triggered by leading edge of 'clk_sys'
11 Path Groups: {reg2reg}
12 Analysis View: func_view_wc
13 Other End Arrival Time 0.000
14 - Setup 0.164
15 + Phase Shift 10.000
16 + CPFR Adjustment 0.000
17 - Uncertainty 0.100
18 = Required Time 0.736
19 - Arrival Time 0.954
20 = Slack Time 0.288
21 Clock Rise Edge 0.000
22 + Clock Network Latency (Ideal) -0.300
23 = Beginpoint Arrival Time -0.300
24
25 Pin Cell Fanout Load Slew Delay Arrival Time Pin Location
26
27 -----
28 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15_reg/CLK -> 4983 0.000 0.200 - -0.300 (1188.00, 635.88)
29 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15_reg/Q sg13p2_dfrbp_1 1 0.011 0.089 0.435 0.135 (1107.60, 636.72)
30 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_inv_4 4 0.044 0.075 0.999 0.234 (1199.04, 636.72)
31 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_inv_8 2 0.062 0.067 0.880 0.315 (1220.16, 661.08)
32 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_inv_16 4 0.088 0.047 0.867 0.381 (1263.84, 662.34)
33 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_inv_8 2 0.032 0.040 0.857 0.438 (1316.04, 683.76)
34 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_inv_8 2 0.030 0.025 0.841 0.470 (1322.37, 706.44)
35 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_inv_2 8 0.104 0.325 0.248 0.718 (1305.12, 757.68)
36 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_inv_2 1 0.018 0.111 0.172 0.800 (1308.24, 751.80)
37 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_inv_4 6 0.025 0.072 0.866 0.806 (1309.68, 711.60)
38 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_mux4_1 1 0.004 0.068 0.301 1.347 (1392.48, 712.32)
39 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_mux4_1 1 0.007 0.069 0.881 1.429 (1393.92, 711.90)
40 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_mux2_2 1 0.007 0.057 0.892 1.521 (1398.24, 718.20)
41 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_mux2_2 1 0.013 0.117 0.133 1.654 (1396.32, 721.56)
42 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_mux2_1 1 0.008 0.164 0.188 1.842 (1331.52, 718.20)
43 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_buf_4 1 0.060 0.066 0.107 2.040 (1334.88, 710.88)
44 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_mux2_2 1 0.006 0.047 0.467 2.107 (1341.92, 806.88)
45 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_mux2_2 1 0.011 0.081 0.891 2.198 (1139.04, 806.88)
46 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_allof_2 1 0.011 0.112 0.138 2.336 (1111.20, 803.10)
47 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_inv_4 6 0.057 0.096 0.122 2.459 (1107.84, 803.52)
48 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_mux2_2 2 0.007 0.079 0.108 2.567 (981.60, 803.52)
49
50 "i:/pr/report_timing_1.rpt" 1061, 177088
```

- Up/DownSize cell cho path timing này MET setup

```
1 #####
2 # Generated by: Cadence Innovus 20.10-p004.1
3 # OS: Linux x86_64 (Host ID: lctc-eda-be-9-ldag-1)
4 # Generated on: Tue Aug 4 22:15:36 2026
5 # Design: croc_chip
6 # Command: report_timing -from i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15_reg/Q -to i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_IFIFOWen_366_reg -report_timing_2.rpt
7 #####
8 Path 1: MET Setup Check with Pin i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_IFIFOWen_366_reg/CLK
9 Endpoint: i_croc_soc/i_croc/i_uart/i_apb_uart_UART_TXFF_IFIFOWen_366_reg/D (*) checked with leading edge of 'clk_sys'
10 Beginpoint: i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15_reg/Q (*) triggered by leading edge of 'clk_sys'
11 Path Groups: {reg2reg}
12 Analysis View: func_view_wc
13 Other End Arrival Time 0.000
14 - Setup 0.164
15 + Phase Shift 10.000
16 + CPFR adjustment 0.000
17 - Uncertainty 0.100
18 = Required Time 0.736
19 - Arrival Time 0.732
20 = Slack Time 0.005
21 Clock Rise Edge 0.000
22 + Clock Network Latency (Ideal) -0.300
23 = Beginpoint Arrival Time -0.300
24
25 Pin Cell Fanout Load Slew Delay Arrival Time Pin Location
26
27 -----
28 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15_reg/CLK -> 4983 0.000 0.200 - -0.300 (1188.00, 635.88)
29 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15_reg/Q sg13p2_dfrbp_1 1 0.011 0.089 0.435 0.135 (1107.60, 636.72)
30 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_inv_4 4 0.044 0.075 0.999 0.234 (1199.04, 636.72)
31 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_inv_8 2 0.062 0.067 0.880 0.315 (1220.16, 661.08)
32 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_inv_16 4 0.088 0.047 0.867 0.381 (1263.84, 662.34)
33 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_inv_8 2 0.032 0.040 0.857 0.438 (1316.04, 683.76)
34 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_inv_8 2 0.024 0.028 0.844 0.482 (1322.32, 706.44)
35 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_inv_4 8 0.104 0.169 0.134 0.616 (1305.12, 757.68)
36 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_inv_2 1 0.016 0.075 0.110 0.724 (1335.36, 765.24)
37 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_inv_4 3 0.022 0.053 0.670 0.804 (1336.32, 705.48)
38 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_buf_1 6 0.024 0.161 0.199 1.003 (1338.44, 797.24)
39 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_mux2_1 1 0.005 0.068 0.183 1.497 (1325.76, 820.20)
40 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_mux2_1 1 0.004 0.060 0.102 1.599 (1325.28, 820.40)
41 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_mux2_1 1 0.005 0.138 0.088 1.713 (1320.12, 705.48)
42 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_mux2_1 1 0.008 0.131 0.181 1.394 (1327.08, 801.36)
43 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_mux2_1 1 0.005 0.068 0.183 1.497 (1325.76, 820.20)
44 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_mux2_1 1 0.004 0.060 0.102 1.599 (1325.28, 820.40)
45 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_mux2_1 1 0.004 0.068 0.175 1.674 (1322.40, 816.64)
46 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_buf_4 1 0.064 0.122 0.212 1.886 (1324.32, 834.96)
47 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_mux2_1 1 0.008 0.075 0.115 2.001 (1062.24, 922.32)
48 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_mux2_2 1 0.008 0.062 0.069 2.070 (1052.16, 927.36)
49 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15/Y sg13p2_mux2_1 1 0.007 0.087 0.106 2.176 (1039.20, 916.02)
50
51 "i:/pr/report_timing_2.rpt" 1061, 177088
```

- Kết quả sau khi dùng lệnh *timeDesign -preCTS -slackReports*

```

innovus 119> timeDesign -preCTS -slackReports
Start to check current routing status for nets...
All nets are already routed correctly.
End to check current routing status for nets (mem=1481.3M)

-----
timeDesign Summary
-----

Setup views included:
func_view_wc

+-----+-----+-----+-----+-----+-----+-----+-----+
| Setup mode | all | reg2reg | in2reg | reg2out | in2out | default | mem2reg | reg2mem |
+-----+-----+-----+-----+-----+-----+-----+-----+
| WNS (ns): | -0.535 | -0.063 | -0.535 | -0.507 | N/A | 6.796 | 0.013 | 0.275 |
| TNS (ns): | -30.496 | -15.684 | -13.822 | -0.990 | N/A | 0.000 | 0.000 | 0.000 |
| Violating Paths: | 697 | 663 | 32 | 2 | N/A | 0 | 0 | 0 |
| All Paths: | 10633 | 10066 | 261 | 35 | N/A | 2 | 1265 | 278 |
+-----+-----+-----+-----+-----+-----+-----+-----+

+-----+-----+-----+-----+
| | Real | Total |
| DRVs | +-----+-----+
| | Nr nets(terms) | Worst Vio | Nr nets(terms) |
+-----+-----+-----+-----+
| max_cap | 128 (128) | -0.027 | 169 (169) |
| max_tran | 0 (0) | 0.000 | 39 (78) |
| max_fanout | 0 (0) | 0 | 2 (2) |
| max_length | 0 (0) | 0 | 0 (0) |
+-----+-----+-----+-----+

Density: 45.012%
Routing Overflow: 0.13% H and 0.32% V
-----
Reported timing to dir ./timingReports
Total CPU time: 1.86 sec
Total Real time: 2.0 sec
Total Memory Usage: 1483.296875 Mbytes
0
innovus 120> innovus 120> █

```